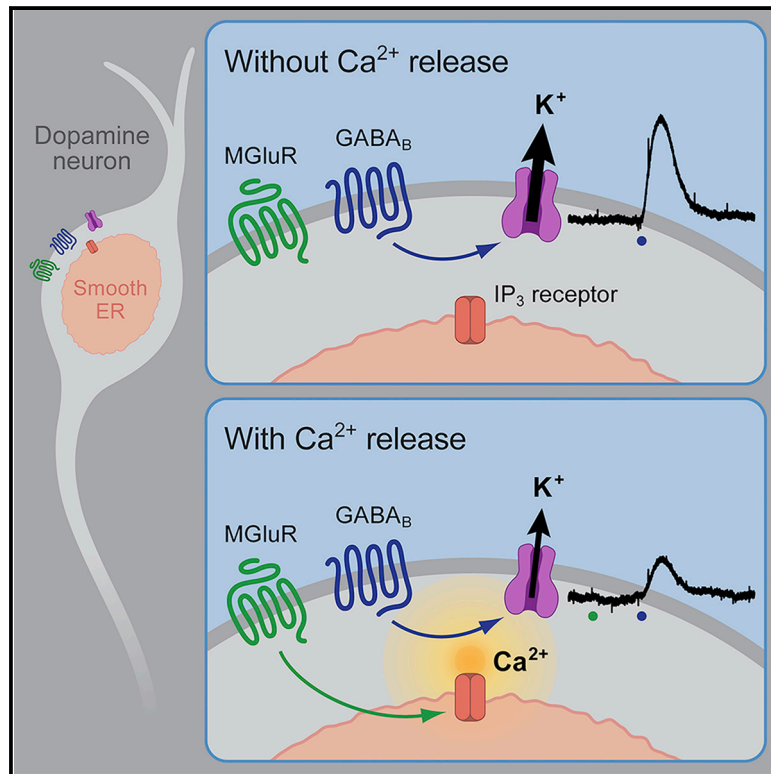


Cell Reports

Calcium Release from Stores Inhibits GIRK

Graphical Abstract



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In Brief

Kramer and Williams show that G_q-mediated calcium release from stores functionally inhibits G_i-coupled GIRK currents. This process occurs within milliseconds and lasts seconds and is not mediated through classical G_q modulators of GIRK such as PKC, PLC, or PLA₂.

Highlights

- mGluR-mediated calcium release from stores inhibits GIRK currents
- GIRK inhibition does not require G_q or G_i receptor activation
- GABA_BR IPSCs are inhibited when calcium is released 1–5 s before stimulation
- Inhibitors of PKC, PLC, PLA₂, and calmodulin had no effect on GIRK inhibition



Calcium Release from Stores Inhibits GIRK

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SUMMARY

Synaptic transmission is mediated by ionotropic and metabotropic receptors that together regulate the rate and pattern of action potential firing. Metabotropic receptors can activate ion channels and modulate other receptors and channels. The present paper examines the interaction between group 1 mGluR-mediated calcium release from stores and GABA_B/D2-mediated GIRK currents in rat dopamine neurons of the Substantia Nigra. Transient activation of mGluRs decreased the GIRK current evoked by GABA_B and D2 receptors, although less efficaciously for D2. The mGluR-induced inhibition of GIRK current peaked in 1 s and recovered to baseline after 5 s. The inhibition was dependent on release of calcium from stores, was larger for transient than for tonic currents, and was unaffected by inhibitors of PLC, PKC, PLA₂, or calmodulin. This inhibition of GABA_B IPSCs through release of calcium from stores is a postsynaptic mechanism that may broadly reduce GIRK-dependent inhibition of many central neurons.

INTRODUCTION

G-protein-coupled receptors (GPCRs) can be separated into three major subclasses defined by their alpha G protein subunit: G_{i/o}, G_{q/11}, or G_s. These divisions confer the receptors with discrete signaling pathways. Dopamine neurons express a variety of GPCRs that activate different ion channels. For example, G_{i/o} GABA_B receptors (GABA_BRs) and dopamine D2 receptors (D2Rs) increase a G-protein-gated inwardly rectifying potassium (GIRK) conductance (Beckstead et al., 2004; Arora et al., 2010). By contrast, G_q group I metabotropic glutamate receptors (mGluR1 and mGluR5) activate the small conductance calcium gated potassium channel SK (Fiorillo and Williams, 1998; Kramer and Williams, 2015). Thus, two signaling pathways activate distinct potassium conductances that hyperpolarize and pause action potential generation in dopamine neurons.

These signaling pathways do not always work in isolation. G_q receptor activation can inhibit GIRK currents generated by G_i receptors, although there are conflicting reports as to the mechanism underlying this interaction. The most commonly proposed mechanism(s) involve protein kinase C (PKC) (Mao et al., 2004). PKC-dependent inhibition has been reported to be either calcium independent (Stevens et al., 1999; Leaney et al., 2001) or

calcium dependent (Hill and Peralta, 2001; Xia et al., 2010). The inhibition of GIRK by phospholipase C (PLC) has also been shown to be directly (Kobrin et al., 2000) and indirectly involved (Keselman et al., 2007). Finally, the activation of phospholipase A₂ (PLA₂), a calcium and G-protein-activated phospholipase, has also been shown to inhibit GIRK (Sohn et al., 2007). Hence, there are a variety of mechanisms underlying the G_q-dependent inhibition of GIRK currents, depending on experimental conditions such as cell type, the particular GPCRs being studied, and agonist application protocols.

To date, the study of inhibition of GIRK by G_q signaling has relied on steady state application of agonists for both G_i and G_q receptors. It is known, however, that group I mGluRs signal differently depending on the method of activation. Steady state application of a selective agonist (DHPG) produced a tonic inward current, whereas transient activation via iontophoresis of aspartate or synaptic release of glutamate produced an outward current (Morikawa et al., 2003; Kramer and Williams, 2015). Thus, distinct intracellular signaling pathways underlie the two methods of activation, and the question of how physiologically relevant receptor activation of group I mGluRs affects GIRK currents remains unexplored.

The present work shows that increases in cytosolic calcium transiently decreased a GIRK conductance that was activated by GABA_BR, D2R, or a non-hydrolyzable GTP analog. Increases in cytosolic calcium were generated by transient mGluR activation or by photolysis of caged-IP₃ or caged-Ca²⁺ loaded into the neuron. The inhibition of GIRK was not affected by inhibition of PLC, PKC, PLA₂, or calmodulin. In addition, the inhibition of transient GIRK currents evoked by synaptic release or iontophoretic application of GABA was greater than steady state currents. The results demonstrate a dynamic role for calcium in the inhibition of GIRK currents in dopamine neurons on the timescale of synaptic transmission, a potential mechanism for inhibition of GIRK currents in other brain areas.

RESULTS

GABA-Receptor-Dependent Decrease in SK Current

To examine the mGluR-induced inhibition of GIRK current, voltage-clamp recordings were made from adult rat Substantia Nigra *pars compacta* dopamine neurons. Iontophoretic application of aspartate at 45-s intervals activated an SK current mediated by mGlu receptors (mGluR1 and mGluR5 (Kramer and Williams, 2015)). All experiments were carried out in the presence of ionotropic glutamate antagonists (see Experimental Procedures). Apamin (100–300 nM) was used to block SK (control

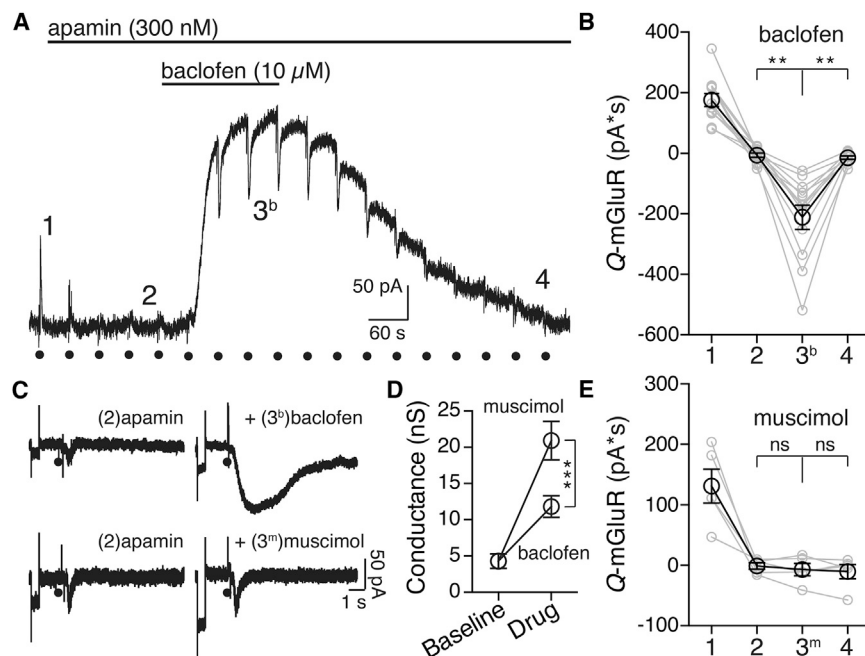


Figure 1. mGluR Activation Induced an Inward Current during GABABR but Not GABAAR Currents

(A) Representative trace of a voltage clamp whole-cell recording showing the block of the SK current by application of apamin (300 nM), followed by application of baclofen (10 μM). The numbers (1, 2, 3, and 4) correspond to time points plotted in (B). Black circles represent the activation of mGluRs with aspartate iontophoresis.

(B) Quantification across cells of the total charge transfer for the mGluR-activated current at baseline (1), after apamin (2) during baclofen (3^b) after washout (4). In the presence of baclofen, aspartate resulted in a ΔGIRK inward current (one-way repeated-measures ANOVA followed by Tukey test, $n = 12$ cells).

(C) Representative episodic traces showing application of aspartate (indicated by the black circle) in apamin (left) showing the small non-selective cation conductance and during either baclofen (black, top) or muscimol (gray, bottom) treatment. Each episode begins with a 3-mV step to assay whole-cell conductance (Figure 1C).

(D) Quantification of the change in cell conductance. Both baclofen and muscimol caused a significant increase in the conductance (two-way repeated-measure ANOVA followed by

Bonferroni, $n = 7$ for muscimol, 11 for baclofen). The increase was significantly larger for muscimol than for baclofen (two-way repeated-measure ANOVA followed by Bonferroni).

(E) Comparison between cells of the aspartate induced current on baclofen or muscimol application. Muscimol and baclofen were not significantly different at baseline (1, $p = 0.94$), after apamin (2, $p > 0.99$), or after washout (4, $p > 0.99$). In muscimol the aspartate-induced current was not different from apamin or washout ($p > 0.99$). All statistics were conducted with a two-way repeated-measures ANOVA followed by a Bonferroni. $n = 5$ (muscimol), $n = 12$ (baclofen), $^{**}p < 0.01$, $^{***}p < 0.001$, bars and summary data points, represent means $\pm$ SEM in (B). Each dot indicates a single cell.

See also Figure S3.

176 ± 21.4 pA*s, in apamin -5.97 ± 6.86 pA*s) in order to isolate the effect of the mGluR activation on the baclofen (10 μM)-induced GIRK current. The small inward current that was induced by aspartate in the presence of apamin resembled the activation of a previously reported non-selective cation conductance (Figure S5E) (Kim et al., 2003; Tozzi et al., 2003).

Application of aspartate in the presence of baclofen and apamin resulted in a separate inward current (-212 ± 40 pA*s; Figure 1A) that was present only during the application of baclofen. That inward current declined as the outward current induced by baclofen reversed upon washing (-14.2 ± 5.3 pA*s; Figures 1A and 1B). The inward current induced by mGluR activation during GIRK activation could result from a breakdown in voltage control secondary to an increase in cell conductance. This mechanism accounted for an inhibition of the hyperpolarization-activated cation current I_h by the activation of GABA_B receptors (Watts et al., 1996). One test of this mechanism was to increase the conductance of the cell with the GABA_A agonist muscimol (10 μM). In the presence of muscimol (10 μM), aspartate iontophoresis did not induce a separate inward current (in apamin -0.99 ± 5.3 pA*s, +muscimol -6.97 ± 10.2 pA*s; Figures 1C and 2E). Thus, the aspartate-induced inward current seen in the presence of baclofen (referred to as ΔGIRK) was not the result of an increase in cell conductance.

When the GABA_BR-dependent current was reduced by the application of BaCl₂ (100 μM), the ΔGIRK current induced by

aspartate also decreased (-206 ± 37 pA*s without BaCl₂ and -57.2 ± 5.1 pA*s with BaCl₂; Figures S3A–S3C). Finally, when mGluR1 and mGluR5 were reversibly inhibited by the low-affinity non-selective antagonist, (S)-α-Methyl-4-carboxyphenylglycine (MCPG) (1 mM), there was a reduction in the inhibition of GIRK (baseline: 64.8% inhibition, in MCPG: 34.5%; Figure S3D). The high-affinity negative allosteric modulators JNJ-16259685 (1 μM, mGluR1) and 2-Methyl-6-(phenylethynyl)pyridine hydrochloride (MPEP) (1 μM, mGluR5) abolished the inhibition of GIRK (inhibition in control: 65.4%, in JNJ/MPEP: 0.9%; Figure S3D). These results suggest that the baclofen induced GIRK current is reduced by transient activation of mGluRs.

mGluR-Dependent Inhibition of GIRK Conductance

The interaction between GABA_B and mGlu receptors was next examined with iontophoretic co-application of GABA and aspartate. This technique allowed activation of GABA_B and mGlu receptors that resembled synaptically stimulated transmitter release. All experiments were carried out in the presence of apamin (300 nM). When aspartate was applied one second before GABA, the peak GABA_B current was reduced to $43.3 \pm 6.7\%$ of control (from 137 ± 12.6 pA to 56.4 ± 9.9 pA; Figure 2C). The current induced by transient activation of GABA_B receptors was inhibited significantly more than the steady state current induced by baclofen as measured from the peak of the baclofen current to the peak of the ΔGIRK current elicited by aspartate

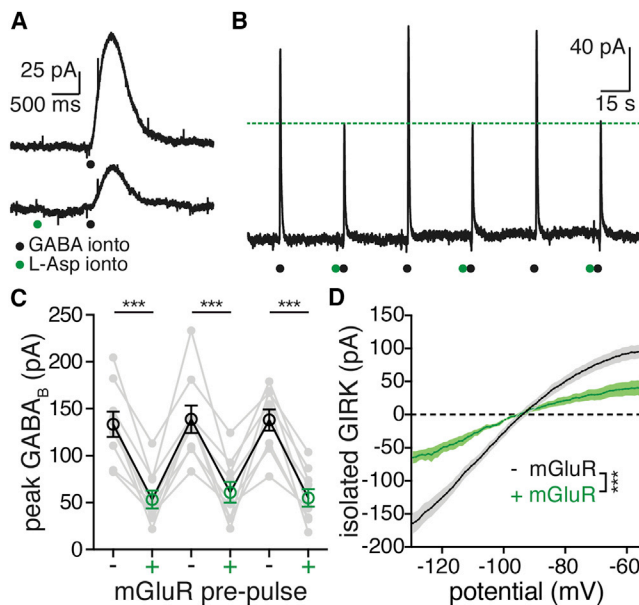


Figure 2. mGluR Activation Decreases GABABR GIRK Currents by Closing GIRK Channels

(A) Representative traces (average of three raw traces for each condition) showing the effect of an mGluR pre-pulse (green dots) on the GIRK current mediated by GABA_B receptor activation (black dots).

(B) Representative chart record from a whole-cell recording showing iontophoresis of GABA (black circles) every 45 s. Aspartate (green circle) was applied by iontophoresis 1 s before every other application of GABA. These experiments were done in the presence of apamin (100–300 nM) as well as GABA_A and AMPA receptor blockers (see [Experimental Procedures](#)).

(C) Grouped data across cells showing the effect of an mGluR pre-pulse (green) 1 s before GABA on the peak GABA_B GIRK current. In black are grouped traces where there was no mGluR pre-pulse. Each gray dot represents an individual cell, black and green circles are means ± SEM. Statistics were performed with a one-way repeated-measures ANOVA followed by a Bonferroni test, $n = 8$ cells, $***p \leq 0.001$.

(D) I - V plots of the GABA_B-mediated GIRK current grouped across cells with (green) and without (black) pre-application of aspartate. Each dot on the I - V plot represents the mean, shaded area around the curves represents SEM. Statistics were performed with a one-way repeated-measures ANOVA, $n = 6$ cells, $***p < 0.001$ (interaction between voltage and mGluR pre-pulse). See also [Figure S5](#).

iontophoresis (inhibition of transient GIRK: $58.5 \pm 5.8\%$, $n = 9$; inhibition of steady state GIRK: $25.6 \pm 3.0\%$, $n = 30$, $p < 0.001$, one-way ANOVA followed by Tukey test).

The voltage dependence of the decrease in the GABA_BR-induced current was examined using a voltage ramp protocol ([Figure 2D](#)). A hyperpolarizing ramp (−55 to −130 mV, 200 ms) was applied just before and at the peak of the GABA_BR outward current (see [Figure S1](#) for detailed protocol). The I - V plot for the GABA_BR current without an mGluR pre-pulse had the characteristic inward rectification of a GIRK conductance and reversed near the potassium reversal potential (-93.7 ± 1.4 mV, calculated $E_K = -103$ mV, not corrected for junction potential; [Figure 2D](#)). When aspartate was applied 1 s prior to GABA, the amplitude of the outward current was reduced, but the rectification and reversal potential were unchanged ([Figure 2D](#); reversal = -94.7 ± 1.2 mV, $p = 0.11$). The inhibition of GIRK was not voltage

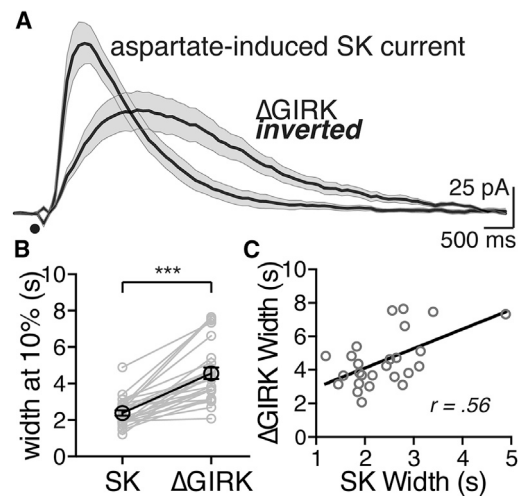


Figure 3. The mGluR-Mediated SK Current and Δ GIRK Inward Current Have Different Kinetics and Are Correlated

(A) Averaged traces of currents (SK and Δ GIRK) following iontophoresis of aspartate (black circle). Each trace is an average of 12 cells, and the shaded area around the mean value represents SEM. Δ GIRK current is inverted to compare the timescale to SK.

(B) Within-cell comparison of the width at 10% of the peak for the SK current and the corresponding Δ GIRK current for that cell. Δ GIRK was significantly longer at 10% of the peak than SK. $***p < 0.001$, statistics were performed with a two-tail paired t test.

(C) Comparing within-cell the width of the SK versus the width (both at 10%) of the Δ GIRK reveals a significant positive correlation, analyzed with a Pearson correlation test, $p < 0.01$, $r = 0.56$, $n = 26$ cells for (B) and (C).

dependent; the scaled I - V plots with and without the mGluR pre-pulse were not different ([Figures S5A and S5B](#)).

Similar results were obtained in experiments using steady state application of baclofen ([Figures S5C–S5E](#)).

Taken together, mGluR activation during, or immediately preceding, a GABA_BR-mediated GIRK current decreased the GIRK conductance. Both the SK current and the Δ GIRK current were mediated by mGluRs; however, the kinetics of the two currents were different. The duration of the SK current was about half that of the inhibition of the GIRK conductance measured at 10% of the peak (SK: 2.38 ± 0.15 s; Δ GIRK: 4.56 ± 0.31 s; [Figures 3A and 3B](#)). But when the duration of each was compared within cells there was a significant positive correlation suggesting an association between the two signaling pathways ([Figure 3C](#)).

Calcium Release from Stores Inhibits GIRK

The mGluR-dependent SK current is activated by release of calcium from stores, and depletion of stores by inhibition of the smooth endoplasmic reticulum calcium pump with cyclopiazonic acid (CPA, 10 μ M) blocks this current ([Morikawa et al., 2000](#)). CPA was therefore tested on the mGluR inhibition of the GIRK conductance. CPA eliminated the SK current and blocked the inhibition of GIRK (aspartate evoked SK in control: 160 ± 27.7 pA*s, after CPA: -18.7 ± 8.49 pA*s, in baclofen: -16.9 ± 11.3 pA*s, wash baclofen: -22 ± 17.4 pA*s; [Figures 4A and 4B](#)).

Strong calcium buffering also blocked the inhibition of GIRK. The control internal solution contained EGTA (100 μ M) that

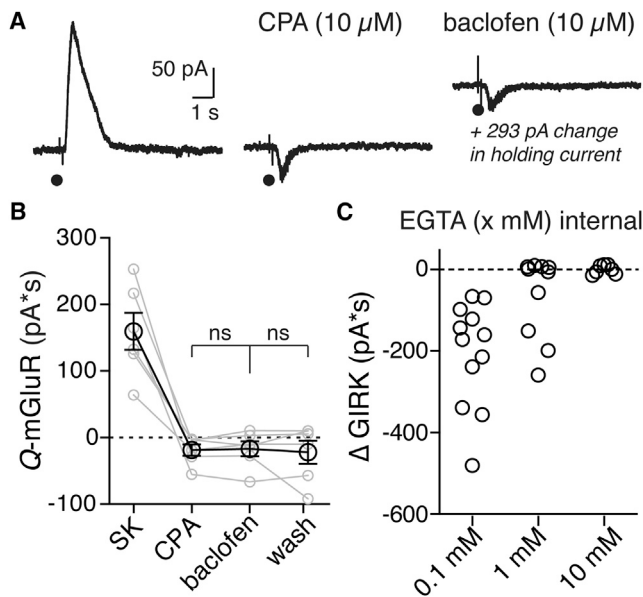


Figure 4. Inhibition of GIRK Requires Calcium Release from Stores

(A) Representative traces showing the currents elicited by aspartate iontophoresis at baseline (left), after CPA (10 μ M, middle), and during baclofen (10 μ M, right). In this example, baclofen elicited a standing outward current of 293 pA (not pictured to scale).

(B) Grouped data showing the aspartate induced current at baseline, after CPA, during baclofen and after washout. Treatment with CPA blocked the aspartate induced a Δ GIRK inward current during baclofen. Each gray dot represents a single cell; lines connect treatments within a cell. Black circles are means $\pm$ SEM. Statistics were performed with a repeated-measures two-way ANOVA followed by a Bonferroni test, ns = not significant ($p > 0.99$), $n = 6$ cells.

(C) Comparison of the Δ GIRK charge during baclofen (10 μ M) with different concentration of EGTA in the internal solution. In EGTA (0.1 mM), there is always a Δ GIRK current (12/12 cells). In 1 mM EGTA, there is a Δ GIRK current in 4/10 cells. In 10 mM EGTA, the Δ GIRK (0/7 cells) was completely blocked. See also Figures S4 and S6.

minimally buffered calcium. The activation of mGluRs with this internal solution always resulted in the activation of SK and an inhibition GIRK (Figure 4C). When the concentration of EGTA was increased to 1 mM, the SK current was inhibited in six of ten cells, and in those six cells, the mGluR inhibition of GIRK was prevented. In the remaining four cells where the SK current persisted, so did the mGluR induced GIRK inhibition (Figure S6B). Increasing the concentration of EGTA to 10 mM blocked both the SK current and inhibition of GIRK after mGluR activation in every cell tested (Figures 4C, S6A, and S6C).

IP₃-Mediated Calcium Release from Stores Inhibits GIRK

To test whether bypassing G protein activation and directly releasing calcium from internal stores could inhibit GIRK, neurons were loaded with caged-IP₃ (100 μ M) that was photolysed by blue light (100 ms, 405 nm). Photolytic release of IP₃ produced an SK current (183 ± 25.7 pA*s, time to 5% of peak: 81.5 ± 10.1 ms, 10%–90% rise time: 49.4 ± 3.6 ms, width at 20% of peak: 677 ± 68.3 ms; Figure 5A), which was blocked by apamin (300 nM, 21.0 pA*s $\pm$ 4.3 pA*s; Figure 5B). When

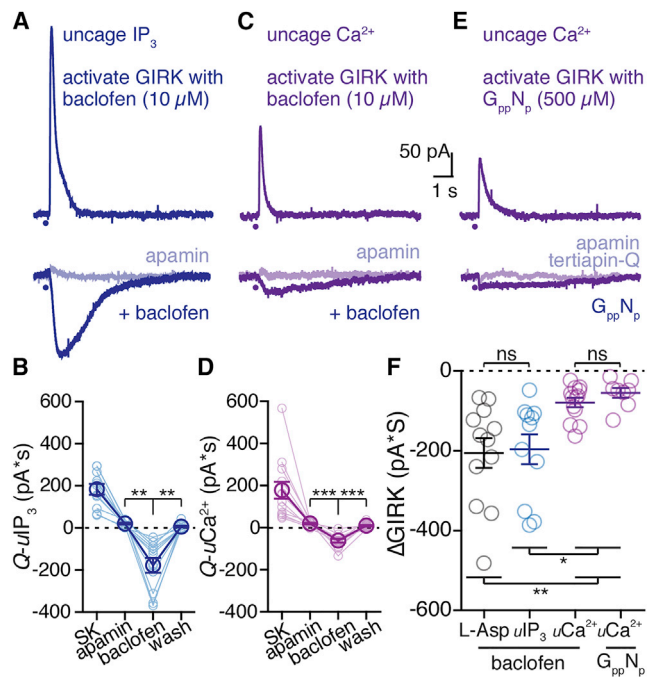


Figure 5. Receptor Activation Is Not Required for GIRK Inhibition

(A) Representative traces from a whole cell voltage clamp recording using photolytic release (405 nm LED pulse) of caged IP₃ (100 μ M) in control (top trace), after application of apamin (bottom light blue trace) and following superfusion of baclofen (10 μ M, blue trace).

(B) Quantified data across cells of all caged-IP₃ flash photolysis experiments. Light blue traces represent data from a single cells; dark blue represents mean data $\pm$ SEM. Statistics performed with a one-way repeated-measures ANOVA followed by a Bonferroni, ** $p < 0.01$, $n = 11$ cells.

(C) Representative traces from a whole-cell voltage clamp recording using photolytic release (365 nm LED pulse) of calcium with DMNPE-4 (1 mM). Top trace shows direct activation of SK, bottom trace (light purple) after apamin, and bottom trace (purple) following superfusion of baclofen (10 μ M).

(D) Quantified data across all caged-Ca²⁺ photolysis experiments. Light purple connected circles represent a data from a single cell; dark purple represents mean averaged data $\pm$ SEM. Statistics performed with a one-way repeated-measures ANOVA followed by a Bonferroni test, *** $p < 0.001$, $n = 13$.

(E) Representative traces from a whole-cell voltage clamp recording using photolytic release of calcium with DMNPE-4. The internal solution also contained GppNHP (GppNp, 500 μ M) that activated GIRK. Top trace is the SK current induced by photolysis of DMNPE-4. Bottom trace (light purple) is following treatment with apamin. Bottom trace is the decrease in GIRK induced by calcium at the peak of the outward current induced by GppNHP.

(F) Comparing the Δ GIRK charge transfer (baseline subtracted) for GIRK currents activated with baclofen (10 μ M) and calcium released via aspartate (black), photolysis of caged-IP₃ (blue), or photolysis of DMNPE-4 (purple, left), and for GIRK currents activated with GppNHP and calcium released by photolysis of DMNPE-4 (purple, right). Each circle represents a single cell; bars, mean value $\pm$ SEM. ns = not significant. Statistics performed with a one-way ANOVA followed by a Tukey test. * $p < 0.05$, ** $p < 0.01$, $n = 12$ (ionto), $n = 11$ (IP₃), $n = 13$ (DMNPE-4, baclofen), $n = 8$ (DMNPE-4, GppNHP) cells.

baclofen (10 μ M, average peak: 425 ± 42.6 pA) was superfused, the photolytic release of IP₃ caused a Δ GIRK current (-177 ± 35.7 pA*s, time to 5% of peak: 143 ± 22.6 ms, 10%–90% rise time: 324 ± 23.9 ms, width at 20% of peak: 2432 ± 181 ms; Figures 5A and 5B). The size of the Δ GIRK current induced by the

release of IP₃ (-196 ± 37 ; Figure 5F) was similar to that induced by aspartate (-206 ± 37 pA*s). The IP₃-dependent inhibition of GIRK was eliminated by depleting calcium stores using CPA (inhibition after CPA: 14.8 ± 5.6 pA*s; in baclofen: -4.3 ± 1.2 pA*s; $p = 0.23$, $n = 4$ cells, one-way repeated-measures ANOVA followed by a Tukey test).

Photolytic Release of Caged-Ca²⁺ Inhibits GIRK

To test whether the photolytic release of calcium alone is sufficient to inhibit GIRK, neurons were dialyzed with a calcium cage (dimethoxynitrophenyl-EGTA (DMNPE-4), see [Experimental Procedures](#)) loaded 90% with CaCl₂. Under these conditions, the upper limit of resting calcium levels in the cell was calculated to be 406 nM and the lower limit 115 nM (see [Experimental Procedures](#)). Calcium was released with UV light (10–100 ms, 365 nm), resulting in the activation of an apamin-sensitive SK current (Figure 5C). Baclofen (10 μ M, 305 ± 28.7 pA) was applied following treatment with apamin, and photolysis of caged-Ca²⁺ produced a Δ GIRK current during baclofen superfusion (calcium current at baseline: 178 ± 40 pA*s; in apamin: 19.7 ± 3.8 pA*s; in baclofen: -61 ± 8.9 pA*s; after wash: 9.4 ± 3.5 pA*s; Figure 5D). Thus, photolytic release of calcium alone results in the inhibition of the GIRK conductance.

To test whether calcium could inhibit GIRK independently of receptor activation, neurons were loaded with a combination of DMNPE-4 caged-Ca²⁺ and guanosine-5'-[(β , γ)-imido] triphosphate (GppNHp) (500 μ M), a non-hydrolyzable GTP analog. Apamin (300 nM) was superfused immediately after the beginning of the recording, and experiments were performed in blockers of GABA_B, mGlu, and D2 receptors. GppNHp produced an outward current that was partially reversed with the GIRK channel inhibitor tertiapin-Q (TPN-Q, 300 nM). The GppNHp-evoked GIRK current (230 ± 17.7 pA) was obtained by subtracting the TPN-Q-sensitive current from the peak outward current. Calcium was photolytically released at the beginning of the recording, at the peak of the GppNHp current, and again after TPN-Q. There was an inward current observed at the peak of the GppNHp current that was absent after TPN-Q treatment, representing a Δ GIRK current (Figure 5E). GIRK currents activated by GppNHp or by baclofen were similarly inhibited by photolytic release of caged-Ca²⁺ (Figure 5F). Thus, the calcium-mediated inhibition of GIRK does not depend on G_i-coupled receptor activation.

Inhibiting Calcium-Activated Enzymes Did Not Block the Inhibition of GIRK

To test the possibility of downstream calcium-activated molecules being required for the inhibition of GIRK, known inhibitors of calcium-activated proteins were tested. Slices were incubated at least 10 min in staurosporine (1 μ M) to inhibit PKC, but the mGluR inhibition of GIRK persisted (apamin: 11.6 ± 7.04 pA*s; baclofen: -179 ± 54.1 pA*s; wash: -14.3 ± 6.69 pA*s; Figure S4A). The PKC inhibitor calphostin-C was also tested but did not block mGluR-mediated GIRK inhibition (apamin: 13.3 ± 14.8 pA*s; baclofen: -505 ± 155 pA*s; wash: -8.12 ± 13.5 pA*s; Figure S4B). Slices were incubated in 4-(4-Octadecylphenyl)-4-oxobutenoic acid (OBAA) (3–10 μ M) to inhibit PLA₂, but the mGluR-mediated GIRK

inhibition was not blocked (apamin: 6.35 ± 15.0 pA*s; baclofen: -397 ± 112 pA*s; wash: -46.9 ± 36.8 pA*s; Figure S4D). The PLC inhibitor U73122 (1 μ M) was included in the pipette and allowed to dialyze into the neuron for at least 10 min, but the mGluR-mediated GIRK inhibition was not affected (peak *I*-GABA at 1 min: $-$ mGluR: 114 ± 19.8 pA, $+$ mGluR: 36.1 ± 7.23 pA; at 7 min: $-$ mGluR: 112 ± 18.7 pA, $+$ mGluR: 31.5 ± 11.5 pA; Figure S4E). Finally, slices were incubated at least 10 min in the calmodulin inhibitor calmidazolium (10 μ M), but this did not block the mGluR-mediated GIRK inhibition (apamin: -10.8 ± 32.6 pA*s; baclofen: -276 ± 25.9 pA*s; wash: -37.3 ± 32.4 pA*s; Figure S4C). The only consistent and robust block of the inhibition of GIRK by mGluR activation is through robust calcium buffering in the neuron or prevention of calcium release from stores.

GABA_B IPSCs Are Inhibited by Calcium

The synaptic release of GABA results in a GABA_B-dependent inhibitory postsynaptic current (IPSC) mediated by GIRK. To test whether calcium inhibits this synaptic current, paired electrically activated GABA_B IPSCs (5 pulses at 100 Hz, 10 s apart) were evoked once per minute. Blockers of mGluR1 and mGluR5 were included in the solution to block synaptic activation of mGluRs by glutamate release. Calcium was released from intracellular stores using photolysis of caged-IP₃ at various times before the second electrical stimulation (Figure 6A). The GABA_B IPSC was maximally inhibited when calcium was released from stores 150–1,000 ms prior to synaptic stimulation (inhibition after 1,000 ms $30.3 \pm 7.7\%$, after 500 ms $33.1 \pm 6.8\%$, after 150 ms $37.8 \pm 7.8\%$; Figure 6B). However, in three of five cells the IPSC was inhibited at last 10% by calcium release from stores 2 s prior to electrical stimulation and in three of eight cells 5 s before stimulation. Thus, a GABA_B IPSC is sensitive to the release of calcium from stores.

Preferential Inhibition of GABA_B-Induced GIRK

Dopamine neurons also express the dopamine D2 receptor (D2R) that activates a GIRK conductance (Beckstead et al., 2004). The inhibition of D2R GIRK current by mGluR activation was examined next. The amplitude of the D2R GIRK currents was smaller than GABA_B GIRK currents. A saturating concentration of quinpirole induced a current that was about the same as that induced by sub-saturating baclofen (baclofen, 1 μ M: 109 ± 17.4 pA; quinpirole, 30 μ M: 138 ± 12.6 pA). Using baclofen (1 μ M) and quinpirole (30 μ M), mGluR activation produced a significantly larger inhibition of the baclofen GIRK current than the quinpirole GIRK current (baclofen control: 2.89 ± 5.78 pA*s, baclofen: -101 ± 19.2 pA*s; quinpirole control: 7.85 ± 6.72 pA*s, quinpirole: -27.9 ± 8.36 pA*s; Figures 7A and 7B).

This experiment was also carried out using iontophoretic application of dopamine, GABA, and aspartate. The application of aspartate 1 s before dopamine significantly and repeatedly reduced the peak dopamine current (from 69.5 ± 7.3 pA to 61.7 ± 6.2 pA; Figure 7D). The percent inhibition of dopamine-mediated GIRK currents by mGluR activation ($11.8 \pm 1.3\%$) was significantly less than the inhibition of GABA-mediated GIRK currents ($56.7 \pm 3.8\%$; Figure 7E).

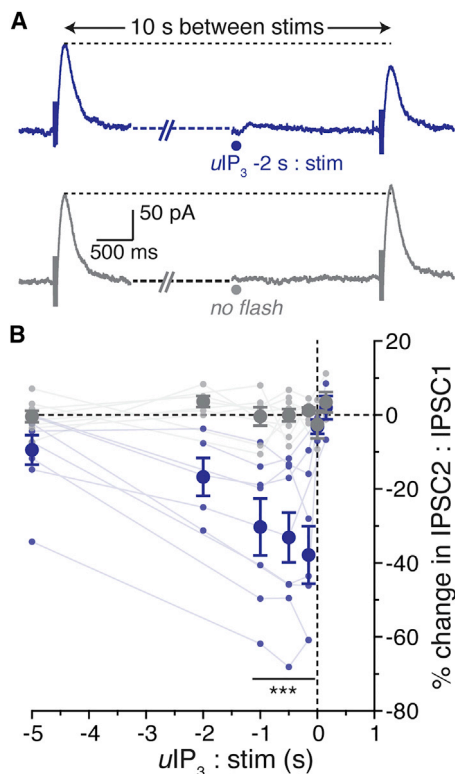


Figure 6. Calcium Release from Stores Inhibits GABAB IPSCs

(A) Representative trace showing GABABR IPSCs evoked with a monopolar stimulating electrode (100 Hz, 5 pulses) in pairs with a 10-s inter-stimulus interval. Photolysis of caged-IP₃ (blue dot, 405 nm, 30–100 ms) was applied at differing times between the pair of stimulations. Experiments were interleaved with traces where IP₃ was not applied (lower trace, gray).

(B) Each small dot represents a data point from a single cell (average of three repetitions), connected by a line to show a within-cell dataset. Points were averaged across cells (large dots). Pairs where IP₃ was released before the second stimulation are in blue; traces with no flash are in gray. Error bars, SEM. Statistics were performed with a two-way repeated-measures ANOVA followed by a Bonferroni test. ****p* < 0.001, *n* = 3–9 cells per averaged data point.

All experiments have been conducted in the presence of apamin to enable the characterization of the selective modulation of the GIRK conductance in the absence of the SK conductance. The interaction of currents induced by aspartate and dopamine or GABA applied using iontophoresis in the absence of apamin was examined next. The currents induced by aspartate, dopamine, or GABA applied alone were measured. Aspartate was then applied 1 s before either dopamine or GABA (Figures S7A and S7B). The linear summation of the aspartate plus GABA or dopamine was compared with the observed current induced by iontophoretic application (GABA: Figures S7A and S7C; dopamine: Figures S7B and S7D). The linear summation of the individual currents evoked by aspartate + GABA or dopamine was always larger than the observed co-application of agonists (GABA summated: 176 ± 27.9 pA, observed: 119 ± 23.7 pA; dopamine summated: 137 ± 20.3 pA, observed: 108 ± 18.2 pA). Furthermore, the inhibition of the peak GABABR current was always significantly greater than for the D2R-mediated GIRK current (Figure S7E).

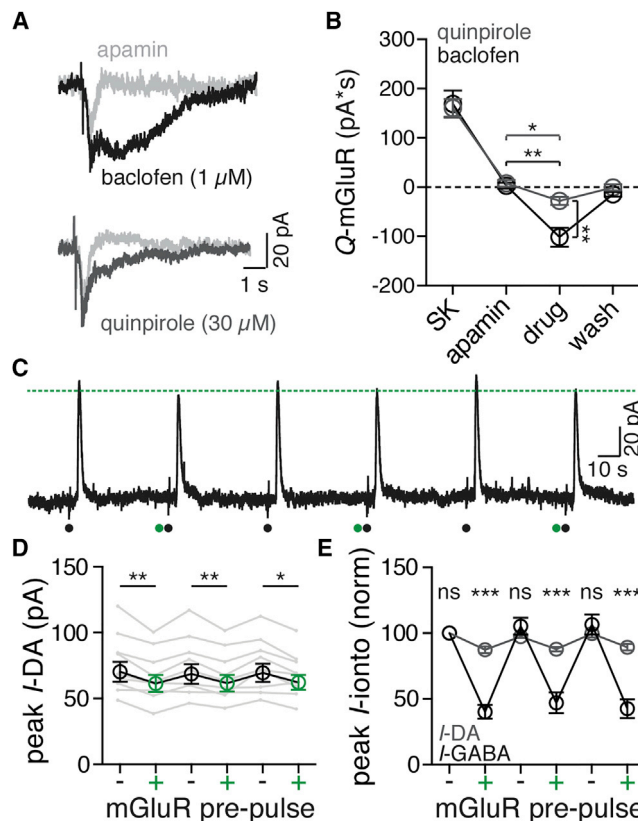


Figure 7. mGluR-Induced Calcium Release Inhibits D2R-Mediated GIRK Currents Less than GABAB

(A) Representative traces of currents induced by iontophoresis of aspartate in the presence of apamin (300 nM, light gray), baclofen (1 μ M, top trace, black), and quinpirole (30 μ M, bottom trace, black).

(B) Quantified data across cells comparing the aspartate induced mGluR currents at baseline (SK), after apamin (300 nM), in either quinpirole (30 μ M, gray) or baclofen (1 μ M, black), and upon washout or reversal of the agonist. Both quinpirole and baclofen resulted in a decrease in GIRK current relative to apamin. Statistics were performed with two one-way ANOVAs followed by a Bonferroni test. **p* < 0.05, ***p* < 0.01, *n* = 8 (quinpirole), *n* = 7 (baclofen); points, means $\pm$ SEM.

(C) Representative trace from a dual iontophoresis protocol with dopamine (black dot) and aspartate (green dot) in the presence of apamin (300 nM). Dotted green line indicates the peak of the GIRK current induced by iontophoretic application of dopamine following application of aspartate. Aspartate iontophoresis caused a consistent decrease the D2R-mediated current.

(D) Grouped data showing the peak amplitude of the dopamine D2-receptor-mediated GIRK current with (green) and without (black) iontophoretic application of aspartate 1 s before dopamine iontophoresis in the presence of apamin (300 nM). Statistics were performed with a one-way repeated-measures ANOVA followed by a Bonferroni test, **p* < 0.05, ***p* < 0.01, *n* = 9 cells. Gray traces indicate within-cell data for single neurons, circles (black and green) $\pm$ SEM.

(E) Grouped data for all dopamine (gray) and GABA (black) dual iontophoresis experiments, normalized to the peak amplitude of the first pulse. Data points indicate means $\pm$ SEM. Statistics were performed with a two-way repeated-measures ANOVA followed by a Bonferroni test, ****p* < 0.001, ns = not significant (*p* > 0.99), *n* = 9 cells for both groups.

See also Figure S7.

DISCUSSION

Transient mGluR activation evoked an IP₃-mediated increase in cytosolic calcium and a decrease GIRK conductance. The inhibition peaked in 1 s and lasted 2–5 s. The activation of mGluRs also inhibited GABA_BR IPSCs. The calcium-mediated inhibition of GIRK was not prevented by pharmacological inhibition of PLC, PKC, PLA₂, or calmodulin, enzymes that had been previously implicated in the G_q-mediated inhibition of GIRK. The only manipulations that blocked the mGluR inhibition of GIRK were depletion of calcium stores and strong cytosolic calcium buffering. Furthermore, direct release of calcium from stores mimicked the GIRK inhibition produced by mGluR activation. Although the precise mechanism remains unclear, the results indicate that there is a functionally significant inhibition of GIRK by calcium released from stores.

Mechanism of GIRK Inhibition by Calcium Release from Stores

The inhibition of GIRK by mGluR activation and photolytic activation of IP₃-mediated calcium release from stores was indistinguishable, but photolytic release of calcium produced a significantly smaller inhibition. This could be because the GIRK was already partially inhibited by the resting level of calcium (between ~115 and ~406 nM), as has been reported (Gantz et al., 2015b). On average, GIRK currents achieved with the same concentration of baclofen were ~30% smaller with caged-Ca²⁺ in the pipette than with caged-IP₃, likely occluding further calcium inhibition and minimizing the apparent effect of photolytic release of calcium. However, despite this constraint, photolytic release of calcium was still able to inhibit GIRK currents whether they were mediated by GABA_BR activation or from inclusion of GppNHP in the pipette.

The most direct interpretation of the present study is that calcium directly inhibits the GIRK channel. It has been reported that external calcium, as well as other divalent cations, can block IRK (including K_{ir} 3.1 and 3.4) channels (Owen et al., 1999). However, there is no complete characterization of any direct inhibition of GIRK by intracellularly positioned calcium ions.

Given what is known about GIRK channel function, there are three potential sites for modulation by calcium. Calcium could act at the Gβγ site to inhibit binding, dissociate Phosphatidylinositol 4,5-bisphosphate (PIP₂) from the channel, or competitively bind to the sodium site. GIRK is an AND gate, meaning Gβγ and PIP₂ are both necessary for channel activation, so negative modulation of either of them would transiently close the channel until calcium was fully sequestered. Sodium, by contrast, enhances the activity of the channel but is not required for current flow, making it a less likely but still possible site of action of calcium (Whorton and MacKinnon, 2011; Wang et al., 2014).

It is possible the inhibition of GIRK currents by calcium is not a direct action of calcium on the channel itself. Although pharmacological inhibitors of PLC, PKC, PLA₂, and calmodulin failed to affect the calcium-mediated inhibition of GIRK, there could be some other unexpected calcium-activated protein responsible for the inhibition of GIRK currents.

IP₃-Mediated SK Currents and GIRK Inhibition Have Different Kinetics

It is of interest to note that, when photolytically releasing caged-IP₃, the SK current onset at 5% of peak was almost twice as fast as the inhibition of GIRK (82 ms versus 143 ms), the SK rise time from 10% to 90% of peak was about 6.5 times faster (49 ms versus 324 ms), and the SK width at 20% of peak was about 3.5 times shorter (677 ms versus 2432 ms). Both processes are calcium dependent, but SK channel activity has been better established and is known to depend solely on intracellular calcium levels and does not desensitize, making it a faithful sensor of intracellular calcium activity (Lancaster et al., 1991). Therefore, the differing kinetics of current onset, time to peak, and width indicate a difference in underlying mechanisms. Without knowing the precise mechanism of calcium-mediated GIRK inhibition, only speculations can be offered to account for the differences in kinetics. For example, the difference in rise time could result from an inherent difference in SK versus GIRK activation or deactivation rates. It also remains possible that an intermediary step is responsible for the inhibition of GIRK.

Preferential GABA_B GIRK Inhibition by Calcium Release from Stores

In the present study, D2R GIRK currents were decreased by calcium release from stores to a significantly smaller degree than the inhibition of GABA_BR GIRK currents. Previous work has shown that GABA_B and D2 receptors have differential sensitivity to calcium. Resting cytosolic calcium levels play a central role in the desensitization and long-term depression of the dopamine D2 receptor but not the GABA_B receptor (Beckstead and Williams, 2007; Gantz et al., 2015b). Increasing cytosolic calcium decreased peak GABA_BR-, but not peak D2R-, mediated GIRK currents (Gantz et al., 2015b).

It is known that D2 receptors associate with the neuronal calcium sensor-1 (NCS-1), which alters desensitization of the receptor (Kabbani et al., 2002). D2 receptors are also localized in close proximity to the calcium channel Ca_v1.3 (Dragicevic et al., 2014). Furthermore, depleting calcium stores or inhibiting calcium entry through the L-type voltage-gated calcium channel increased the amplitude of a D2R-mediated GIRK current in wild-type mice (Gantz et al., 2015b). Given the results of the present study in light of these results, it is possible that D2-receptor-mediated GIRK currents at baseline are nearly maximally inhibited by calcium entry through Ca_v1.3, IP₃ receptors and/or resting calcium levels. Indeed, D2R-mediated currents in the present study were consistently smaller than GABA_BR-mediated currents. A second interpretation of the present results is that calcium release from stores was preferentially affecting GABA_BR GIRK currents because of the close proximity of IP₃ receptors with the GABA_B signaling complex and separated compartmentalization of the D2R complex.

A General Mechanism to Decrease GIRK-Dependent Synaptic Inhibition

Glutamate release resulting in an mGluR-mediated IPSC could have an initial inhibitory SK-mediated component followed by a period lasting as long as 5 s during which GIRK-mediated inhibition could be decreased. Given that some VTA neurons have very little SK-mediated current (Wolfart et al., 2001),

the activation of mGluRs would disinhibit cells through the calcium-mediated inhibition of GIRK-dependent synaptic currents. Furthermore, many neurons have functional GIRK currents and IP₃ receptors, such as hippocampal pyramidal cells (Nakamura et al., 2000; Degro et al., 2015) and 5-hydroxytryptamine (5-HT) neurons of the dorsal raphe (Gantz et al., 2015a; Pan et al., 1994). Thus, the interaction between calcium release from stores and GIRK channel currents described here may not be a dopamine neuron-specific effect.

Finally, calcium buffering and manipulations of the resting level of free calcium modulates the GIRK conductance activated by both D2 and GABA_B receptors in that strong calcium buffering with 1,2-Bis(2-aminophenoxy)ethane-N,N',N',-tetraacetic acid (BAPTA) (10 mM) results in a larger GIRK currents (Beckstead and Williams, 2007; Gantz et al., 2015b). In addition, the regulation of GIRK conductance by calcium is modulated following treatment of animals with psychostimulants. The calcium-dependent inhibition of GIRK is augmented following treatment of animals with cocaine (Arora et al., 2011; Gantz et al., 2015b) and animals that self-administered methamphetamine (Sharpe et al., 2014). The mechanisms underlying these observation may differ in that there was no receptor dependence in animals that self-administered methamphetamine (Sharpe et al., 2014), whereas the present results indicate that the GABA_B-induced currents are more susceptible to the calcium dependent regulation.

To close, GIRK channels are well-known mediators of neuronal inhibition, and the modulation of this conductance has been studied extensively. Typically, modulation has been studied on the order of minutes (Xia et al., 2010) to days (Arora et al., 2011). The data presented here show that release of calcium from stores inhibits GIRK currents in milliseconds and reverses in seconds, a dynamic process that can modulate on the timescale of GIRK-dependent synaptic transmission. This demonstration of the direct effect of calcium on GIRK currents has broad implications for inhibition mediated by GIRK channels.

EXPERIMENTAL PROCEDURES

Animals

All animal experiments were performed in accordance with the NIH guidelines and with approval from the Institutional Animal Care and Use Committee of the Oregon Health and Science University (Portland, OR). Adult (6–10 weeks), male and female, Sprague-Dawley rats were used for all experiments.

Slices and Solutions

Rats were anesthetized in a vapor chamber (isoflurane, OHSU pharmacy) and killed. Brains were extracted and blocked, removing the cerebellum and frontal cortex rostral to the optic chiasm, the dorsal surface of the brain was fixed onto the vibratome stage (krazy glue), and slices (230 μ m) containing the Substantia Nigra were prepared. Slices were cut using a vibratome (Leica) in warm (30°C) physiological artificial cerebrospinal fluid (ACSF) containing (in mM): 126 NaCl, 2.5 KCl, 1.2 NaH₂PO₄, 1.2 MgCl₂, 2.4 CaCl₂, 11 glucose, 21.4 NaHCO₃, and 5 MK801, saturated with 95% O₂ and 5% CO₂ (pH 7.4), 300 mOsm/kg, and then incubated in the same solution warmed to 34°C for 30–60 min. After incubation slices were transferred into the physiological ACSF lacking MK801. Slices were placed in a recording chamber and superfused with warmed (35°C) physiological ACSF at 1.5–2 ml/min.

Whole-Cell Recording

All recordings were performed from dopamine neurons of the Substantia Nigra *pars compacta*, defined as a region lateral of the Medial Terminal Nucleus of

the Accessory Optic Tract (MT). Dopamine neurons were identified by their large cell bodies, the characteristic pacemaker-like firing (1–5 Hz) recorded in the cell-attached mode, and the presence of a large *I_h* current. Whole-cell pipettes had resistances of 1.3–1.8 M Ω . Unless otherwise noted, pipette solutions used for whole-cell recordings contained (in mM): 115 K-methanesulfonate, 20 NaCl, 1.5 MgCl₂, 10 HEPES, 0.1 EGTA, 2 Mg-ATP, 0.25 Na₂-GTP, and 10 Na₂phosphocreatine (pH 7.42), 283 mOsm/kg. Voltage-clamp recordings were made (holding potential –55 mV) using an Axopatch 200B amplifier (Axon Instruments, Foster City, CA). Series resistance was monitored using a 3-mV test pulse (collected at 50 kHz and filtered at 10 kHz). Recordings were discarded if the series resistance exceeded 12 M Ω . Episodic data were obtained at 5 kHz and filtered at 2 or 5 kHz using AxoGraph X (Axon Instruments). Continuous recording at 200 Hz was obtained using Chart (version 5.5.6; AD Instruments, Colorado Springs, CO).

Iontophoretic Application

Transient application of agonists (dopamine, GABA, or aspartate) using iontophoresis produced outward currents that mimicked synaptically evoked IPSCs. Iontophoresis was performed with an Axoclamp 2A amplifier (Axon Instruments) using pipettes containing L-aspartate (1 M [pH 7.4], 40–70 M Ω , –80 to –130 nA ejection, and +1.5–3 nA backing current), dopamine (1 M, 40–60 M Ω , +30 to –100 nA ejection and –1.5 to –3 nA backing current), or GABA (500 mM [pH 7.3], 30–50 M Ω , +100 to –190 nA ejection, and –0.2 to –3 nA backing current). Iontophoretic pipettes were placed within 5–10 μ m of the soma. mGluR currents were isolated after pretreatment with (5S,10R)-(+)-5-methyl-10,11-dihydro-5H-dibenzo[a,d]cyclohepten-5,10-imine (MK-801, 50–100 μ M) and in the presence NBQX (300 nM). Iontophoretic experiments with GABA included picrotoxin (100 μ M) and SR-95531 (300 nM) to block GABA_A receptors. Iontophoretic experiments with dopamine, included prazosin (100 nM) in the superfusion solution to block α 1-adrenoceptors (Paladini et al., 2001). Apamin (100–300 nM) was used to block SK-mediated outward currents while testing the block of GIRK by mGluR activation. The protocol illustrated in Figure S1 was used to test the effect of aspartate iontophoresis on the current elicited by GABA or dopamine.

Synaptic Stimulation

GABA_B-mediated IPSCs were evoked with a train (5 pulses at 100 Hz) using a monopolar electrode (filled with physiological ACSF, 2–2.5 M Ω) placed ~70–100 μ m lateral and ~40–60 μ m caudal from the patched neuron. Stimulation intensity was adjusted to obtain IPSCs (> 20 pA) that did not evoke a direct action current in the recorded neuron. GABA_B IPSCs were isolated by including NBQX (300 nM), picrotoxin (100 μ M), sulpiride (300 nM), JNJ-16259685 (1 μ M), and MPEP (300 nM) to block α -amino-3-hydroxyl-5-methyl-4-isoxazolepropionic acid (AMPA), GABA_A, D2, mGluR1, and mGluR5 receptors, respectively. Slices had been pre-incubated in MK801 (50–100 μ M) to block NMDA receptors. In most experiments, CGP-55845 (300 nM) was applied at the end of the experiment to verify that the IPSC resulted from GABA_B receptors. Photolysis experiments of caged-IP₃ in combination with the GABA_B IPSC included at least three trials per time point that were averaged. IPSCs were evoked at 1-min intervals and interleaved with experiments using the combination of photolysis and electrical stimulation.

Construction of Current-Voltage Plots

Current/voltage (*I/V*) plots were constructed using ramp protocols. Agonist-induced currents were isolated by subtracting the evoked current from its respective baseline. Averages of at least two ramps were combined before subtraction. The ramp protocol was used in combination of agonists applied by superfusion (at steady state; Figure S2) and iontophoresis (transient; Figure S1B).

Caged Compound Photolysis

Photolysis of caged-IP₃ (Walker et al., 1989) and caged-Ca²⁺ (Ellis-Davies and Barsotti, 2006) was performed on a separate experimental setup. Whole-cell recordings were made with an Axopatch 1D amplifier (Molecular Devices, Sunnyvale, CA). Episodic currents were recorded at 10 kHz for 1 min using AxoGraphX (1.4.3; AxoGraphX, Berkeley, CA).

A stock solution of caged-IP₃ (0.1 mM) was added to normal internal. Photolysis of caged-Ca²⁺ was done with a modified internal solution where

DMNPE-4 (1 mM) and CaCl_2 (0.9 mM) were added. All experiments were carried out in the dark.

All photolysis experiments were carried out with full-field illumination (365 or 405 nm LED, Thorlabs, Sterling, VA) coupled through a $\times 60$ objective (Olympus, 0.9 na). The IP_3 generated SK currents were of similar size with 365 or 405 nm LED photolysis. The 365-nm LED was used for photolysis of caged- Ca^{2+} .

Calculating Resting Calcium Levels

The Chelator web application, version 1.3 (maxchelator.stanford.edu), was used to calculate resting calcium levels using the values given for whole-cell voltage clamp experiments, and 1.1 mM buffer with 900 μM added CaCl_2 . Because there are two buffers in the internal that have slightly different affinities for calcium at pH 7.4 (DMNPE-4: 19 nM and EGTA: 67 nM) a range for the possible resting calcium concentration was calculated. The upper estimate was based on the EGTA affinity, whereas the lower limit was based on DMNPE-4.

Drugs

Caged- IP_3 was a generous gift from Dr. Hitoshi Morikawa (University of Texas, Austin, TX), who obtained it from Dr. Kamran Khodakhah (Albert Einstein School of Medicine, Bronx, NY). DMNPE-4 caged- Ca^{2+} was a generous gift from Dr. Graham Ellis-Davies (GM053395, Ichan School of Medicine at Mount Sinai, New York, New York). CGP-55845, JNJ-16259685, and MPEP were obtained from Tocris (Minneapolis, MN). Apamin was obtained from Calbiochem (San Diego, CA). NBQX and MK801 were obtained from Abcam (Cambridge, MA). SR-95531 was obtained from Hello Bio (Princeton, NJ). The remaining compounds were obtained from Sigma-Aldrich (St. Louis, MO).

Data Analysis

To determine the total charge transfer induced by mGluR activation and to account for the duration of GIRK inhibition, the area under the curve (AUC) was measured from the start of the current until the end and expressed as $\text{pA}\cdot\text{s}$. For most experiments, apamin was used to isolate inward currents. When the peak of the evoked current was measured, 30 ms of data were averaged around a center point. Data are expressed as mean $\pm$ SEM. For data in Figure S7, currents were summated using AxoGraph X. Statistical significance was determined with Student's *t* test or one-way ANOVA followed by the Tukey's post hoc test or Bonferroni test. The difference was considered significant at $p < 0.05$. When necessary a two-way ANOVA was performed followed by Bonferroni's post hoc test. Data were graphed and analyzed using Prism 6 (GraphPad, La Jolla, CA). Figures were prepared using Illustrator CS5.1 (Adobe, San Jose, CA). At least three biological replicates were tested per condition.

SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures and can be found with this article online at <http://dx.doi.org/10.1016/j.celrep.2016.11.076>.

AUTHOR CONTRIBUTIONS

Conceptualization, P.F.K. and J.T.W.; Methodology, P.F.K. and J.T.W.; Investigation, P.F.K.; Formal Analysis, P.F.K.; Resources, P.F.K. and J.T.W.; Writing – original draft, P.F.K.; writing – review & editing, P.F.K. and J.T.W.; Visualization, P.F.K.; Supervision, J.T.W. Funding Acquisition, P.F.K. and J.T.W.

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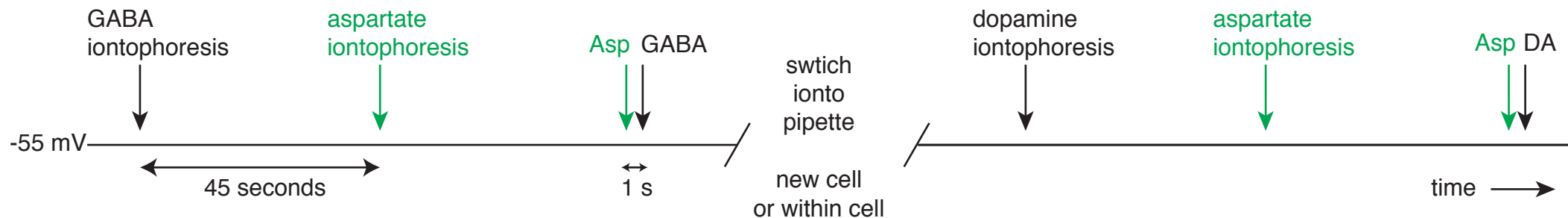
Supplemental Information

Calcium Release from Stores Inhibits GIRK

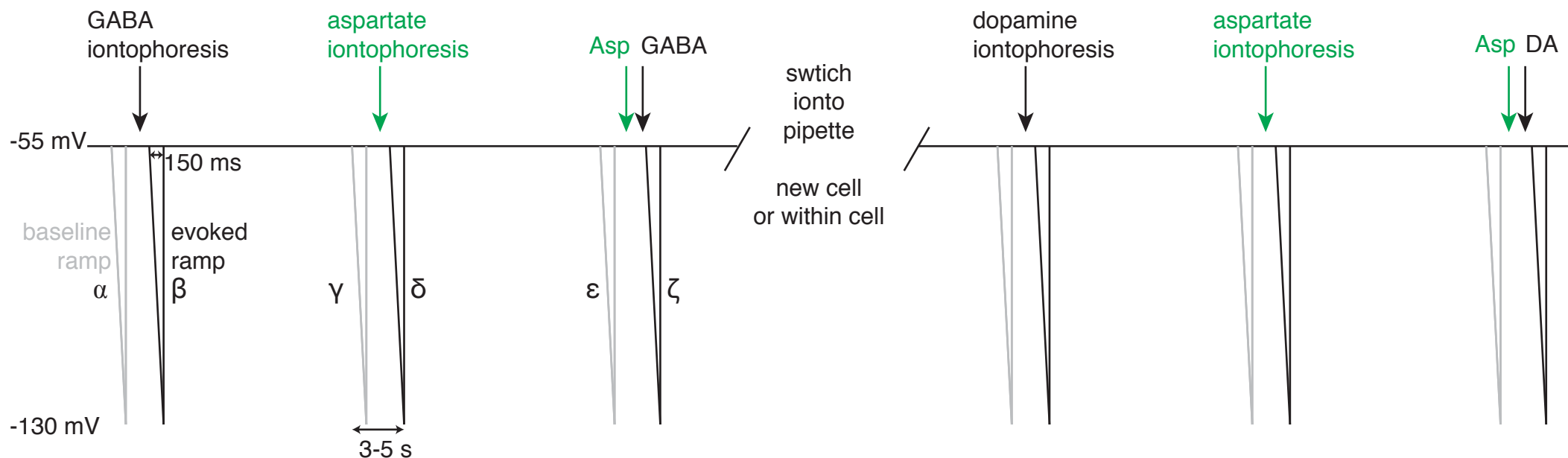
Paul F. Kramer and John T. Williams

A. Dual iontophoresis

performed in apamin (300nM; Figures 2 and 7C,D,E), and without apamin (Figure S7)

**B. Dual iontophoresis with I/V protocol**

always in apamin (300 nM)



$\beta - \alpha$ = isolated GABA_B

$\delta - \gamma$ = isolated mGluR

$\zeta - \epsilon$ = mGluR + GABA_B

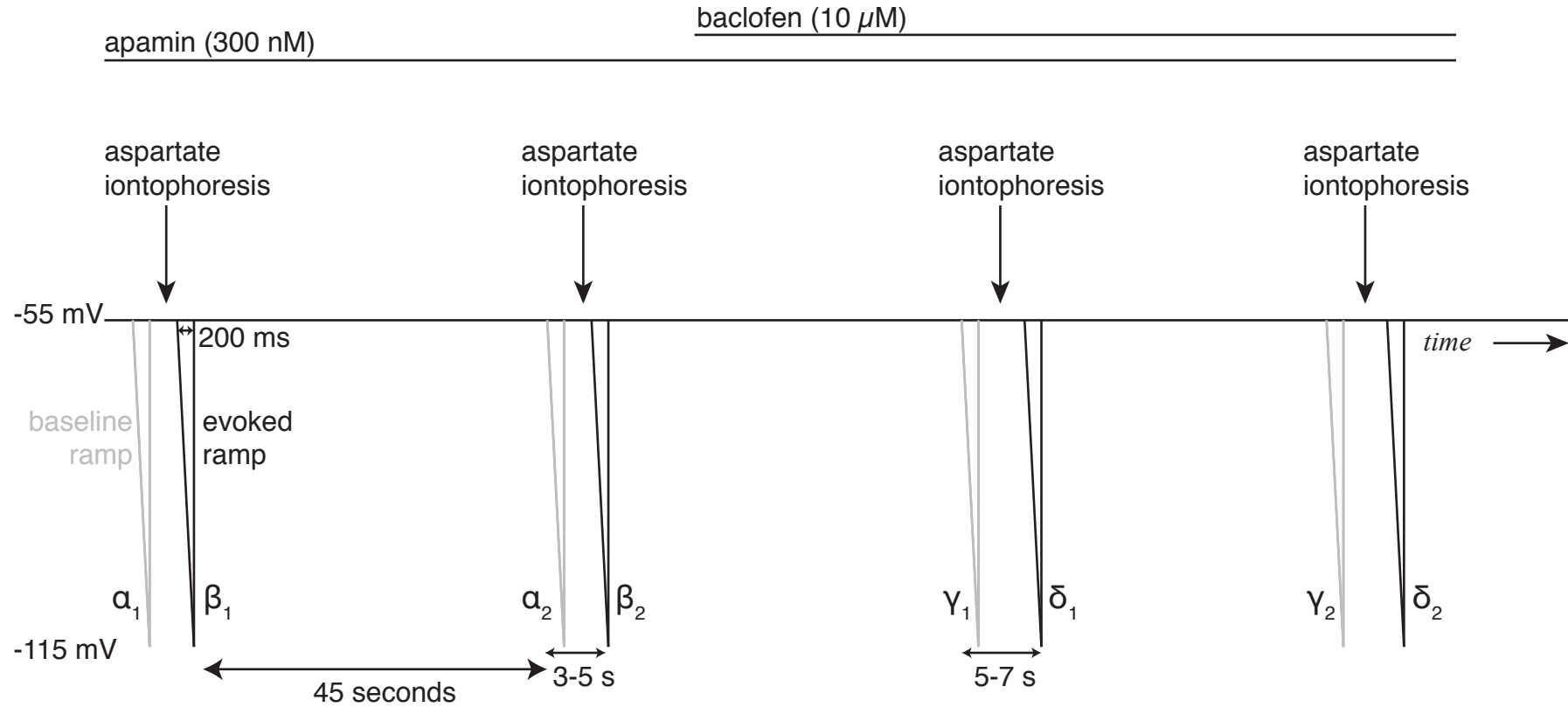
mGluR + GABA_B - isolated mGluR = isolated effect of mGluR on GABA_B

Figure S1 – Related to Experimental Procedures “Iontophoretic Application”. Protocol illustrations for sequential co-iontophoresis with and without voltage ramps.

A. On the first sweep GABA was applied 45 s prior to aspartate, on the next sweep GABA was applied 1s after the application of aspartate. This protocol was repeated periodically. After 3 complete repetitions, the iontophoretic pipette containing GABA was retracted and a pipette filled with dopamine was brought close to the neuron.

B. Same general protocol as in **A**, except voltage ramps were performed as illustrated. The current resulting from these ramps at the peak of the GABA current was then subtracted from the current in control, in the order to isolate the current induced by GABA before and following activation of mGluRs.

A. I/V protocol for steady-state GABA_B activation



$$((\beta_2 + \beta_1) / 2) - ((\alpha_2 + \alpha_1) / 2) = \text{isolated mGluR inward}$$

$$((\delta_2 + \delta_1) / 2) - ((\gamma_2 + \gamma_1) / 2) = \Delta \text{GIRK}$$

$$((\gamma_2 + \gamma_1) / 2) - ((\alpha_2 + \alpha_1) / 2) = \text{isolated baclofen I-GIRK}$$

$$((\delta_2 + \delta_1) / 2) - ((\beta_2 + \beta_1) / 2) = \text{isolated effect of aspartate on baclofen I-GIRK}$$

Figure S2 – Related to Experimental Procedures “Construction of *I*-V plots”. Construction of *I*-V plots for steady-state baclofen voltage ramps.

A. Aspartate was iontophoresed every 45 seconds. Voltage ramps were performed as illustrated in apamin, and again during baclofen (10 μ M) application. The resulting currents were subtracted from their respective baselines to construct isolated *I*-V plots of interest.

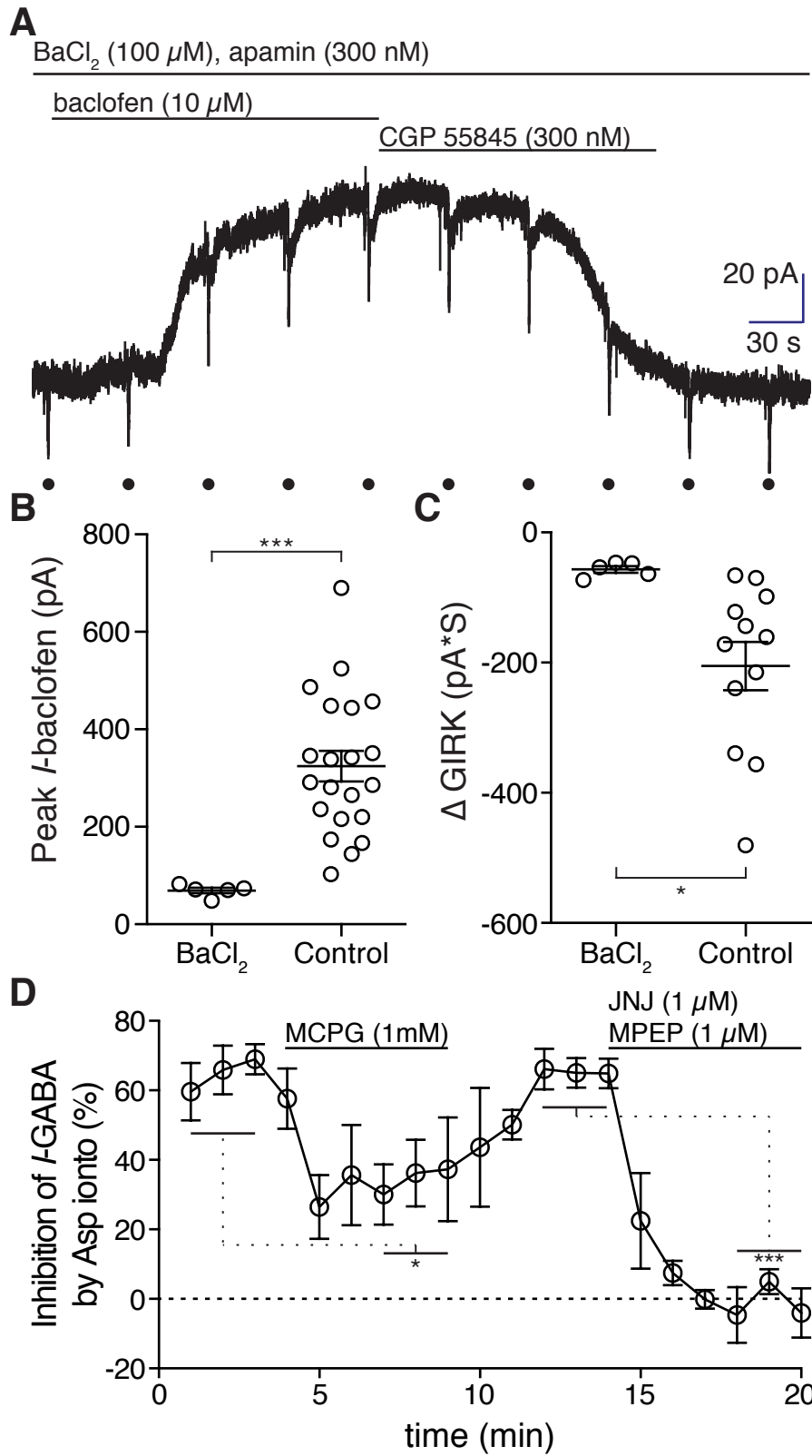


Figure S3 – Related to Figure 1. BaCl₂ decreased baclofen GIRK current and reduced the inward current induced by aspartate.

A. Representative trace from a whole-cell recording showing the reduced baclofen (10 μ M) mediated GIRK current in the presence of BaCl₂ (100 μ M) and apamin (300 nM), and the reduced size of the inward current mediated by aspartate iontophoresis during that current. The baclofen-mediated GIRK current was reversed by the specific GABA_BR antagonist CGP-55845 (300 nM). Black circles represent the activation of mGluRs with aspartate iontophoresis.

B. Quantification across cells of the peak baclofen-mediated GIRK current. BaCl₂ significantly reduced the peak compared to controls.

C. Quantification across cells of the aspartate mediated Δ GIRK inward current during baclofen (10 μ M) application. BaCl₂ significantly reduced the aspartate induced Δ GIRK. Statistics were performed with a 2-tailed unpaired t-test, n = 5 cells (BaCl₂) and 8 cells (control), *p < 0.05, ***p < 0.001, bars represent means and error bars s.e.m.

D. Percent inhibition of GABA_BR-mediated GIRK currents were calculated using the dual iontophoresis method. MCPG (1 mM) was applied after a baseline was achieved, and was washed out. After a second baseline was achieved, JNJ and MPEP (1 μ M each) were applied. Statistics were performed with a 2-way repeated measures ANOVA followed by a Bonferroni post-hoc test. n = 4 cells, *p < 0.05, ***p < 0.001, bars represent means and error bars s.e.m.

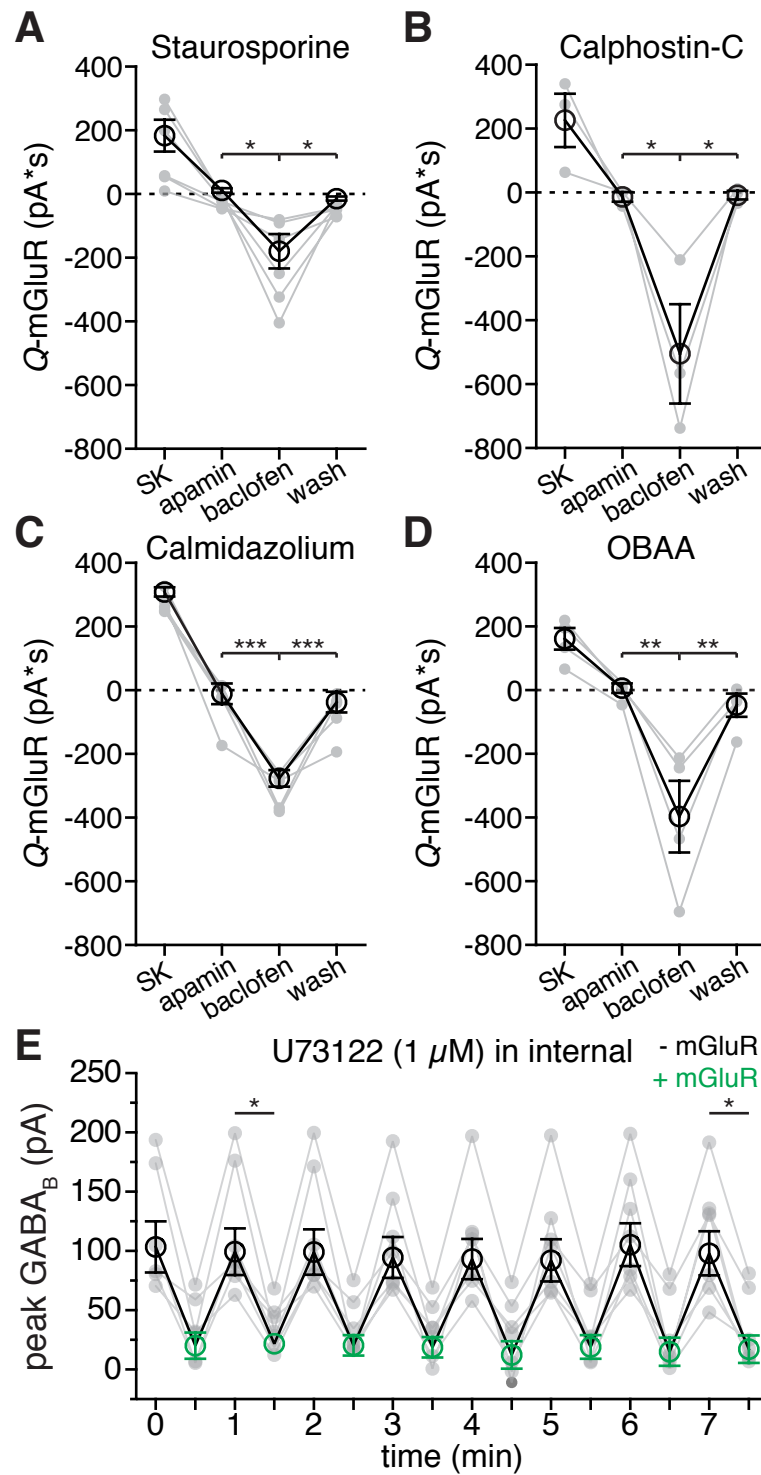


Figure S4 – related to Figure 4, and results “inhibiting calcium-activated enzymes did not block the inhibition of GIRK” – Pharmacological inhibition of calcium-activated lipases and other proteins has no effect on GIRK inhibition.

A. Slices were perfused with staurosporine (1 μ M) for at least 10 minutes. In the presence of staurosporine, mGluR-mediated inhibition of GIRK was not impaired (apamin: 11.6 ± 7.04 pA*s; baclofen: -179 ± 54.1 pA*s; wash: -14.3 ± 6.69 pA*s; n = 6).

B. A second PKC inhibitor, calphostin-C (10 μ M) was also examined. After incubating slices in the vial for an hour and keeping the slices in the drug during recordings, mGluR inhibition of GIRK was not impaired (apamin: 13.3 ± 14.8 pA*s; baclofen: -505 ± 155 pA*s; wash: -8.12 ± 13.5 pA*s; n = 3).

C. The PLA₂ inhibitor OBAA (Köhler et al., 1992) was tested on the mGluR-mediated inhibition of GIRK. Slices were treated with OBAA (3–10 μ M, in the bath for 15 min), or incubated in OBAA (1 μ M) for over an hour before transferring to the recording chamber. mGluR-mediated inhibition of GIRK was not impaired (apamin: 6.35 ± 15.0 pA*s; baclofen: -397 ± 112 pA*s; wash: -46.9 ± 36.8 pA*s; n = 4).

D. The effect of calmodulin was tested by using the inhibitor calmidazolium (10 μ M). In the presence of calmidazolium, mGluR-mediated inhibition of GIRK was not impaired (apamin: -10.8 ± 32.6 pA*s; baclofen: -276 ± 25.9 pA*s; wash: -37.3 ± 32.4 pA*s; n = 5).

E. Neurons were loaded with the PLC inhibitor U73122 (1 μ M) for at least 10 minutes before testing the inhibition of GIRK using the dual iontophoresis method. U73122 did not affect the mGluR-mediated inhibition of GIRK (At 1 minute: - mGluR: 114 ± 19.8 pA, + mGluR: 36.1 ± 7.23 pA; at 7 min: - mGluR: 112 ± 18.7 pA, + mGluR: 31.5 ± 11.5 pA; n = 7) *p<0.05, **p<0.01, ***p<0.001; all analyses performed with a repeated measures one-way ANOVA followed by a Bonferroni post-hoc test.

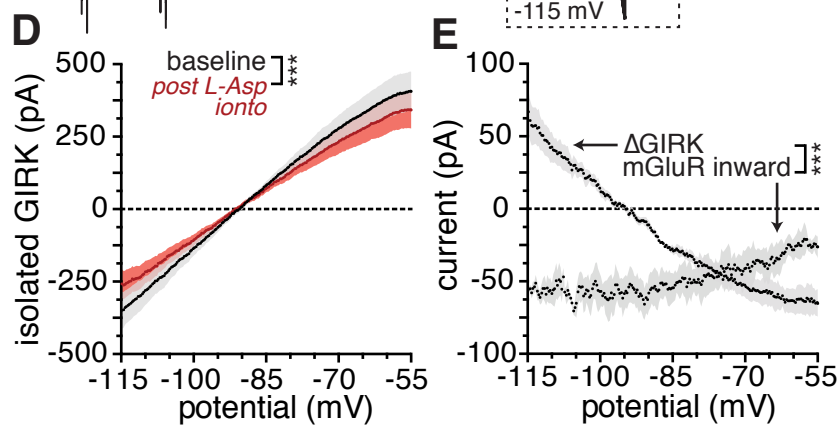
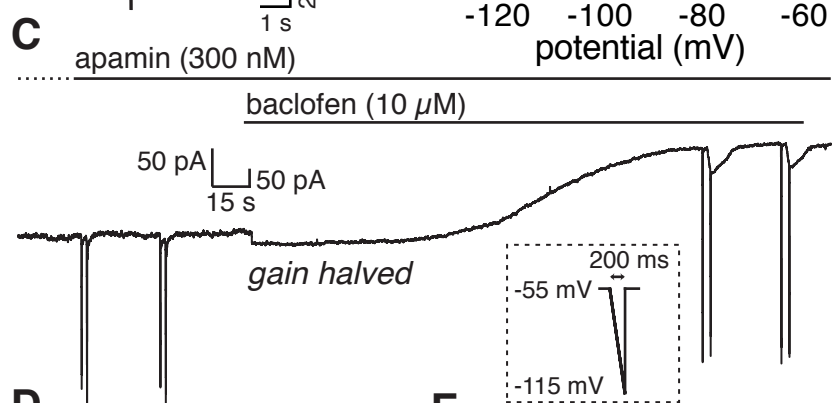
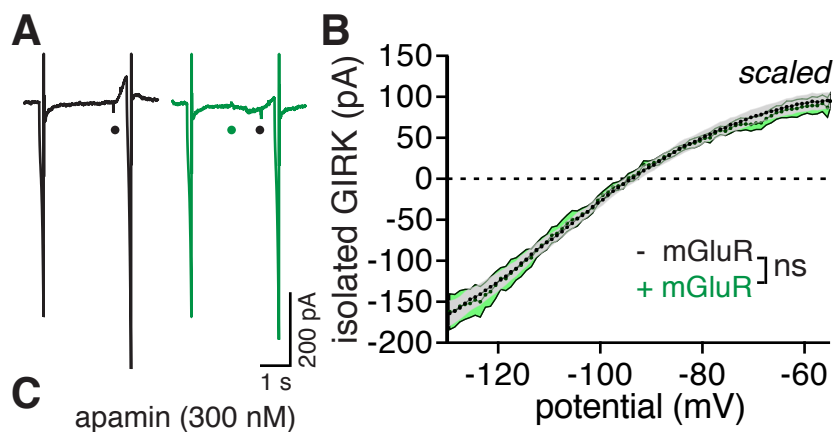


Figure S5 – Related to Figure 2. Block of steady state GIRK currents is similar to transient currents.

A. Representative traces showing the protocol for determining the current–voltage (I – V) relationship of the GABA_BR current with (green) and without (black) an mGluR pre-pulse. See methods for a detailed protocol.

B. Mean isolated GABA_BR GIRK I – V plots with mGluR pre-pulse curve (green) scaled to the curve lacking an mGluR pre-pulse (black). Scaling factor (no pre-pulse / pre-pulse amplitude) was determined for each cell at -55 mV and then applied across all voltages. After scaling, the two curves were no longer significantly different by a one-way repeated measures ANOVA, $n = 6$ cells, $p = 0.94$ (interaction between voltage and mGluR pre-pulse), ns = not significant.

C. Representative whole-cell voltage clamp trace showing generation of isolated I – V plots for the mGluR inward current in apamin, and the Δ GIRK current during baclofen application. *Inset* example voltage ramp used for determining I – V relationships. The amplifier gain (α) was halved from $\times 10$ to $\times 5$ upon application of baclofen to prevent saturation of the signal during voltage ramps. The scale bar reflects the two different vertical scales.

D. Isolated GIRK I – V curves before (black) and after (red) iontophoresis of aspartate, obtained by subtracting the ramp in baclofen from the corresponding ramp before baclofen application (see methods). The decrease in the I – V is significantly different from the control I – V , when using a two-way repeated measures (both factors) ANOVA $n = 6$ cells, $***p < 0.001$ (interaction between voltage and current).

E. Isolated aspartate-induced decrease in GIRK and mGluR inward current I – V curves obtained by subtracting the baseline ramp from the ramp following aspartate iontophoresis, before and during baclofen treatment. The decrease in GIRK reverses near E_k , and the shape resembles what would be predicted for a decrease in GIRK conductance. The mGluR inward I – V ramps agrees with previous reports examining this current and resembles a non-selective cation conductance. These two I – V ramps were significantly different. Two-way repeated measures ANOVA (single factor), $n = 3$ (mGluR inward) $n = 6$ (Δ GIRK), $***p < 0.001$ (main interaction between current and voltage) and $p < 0.01$ main effect of current.

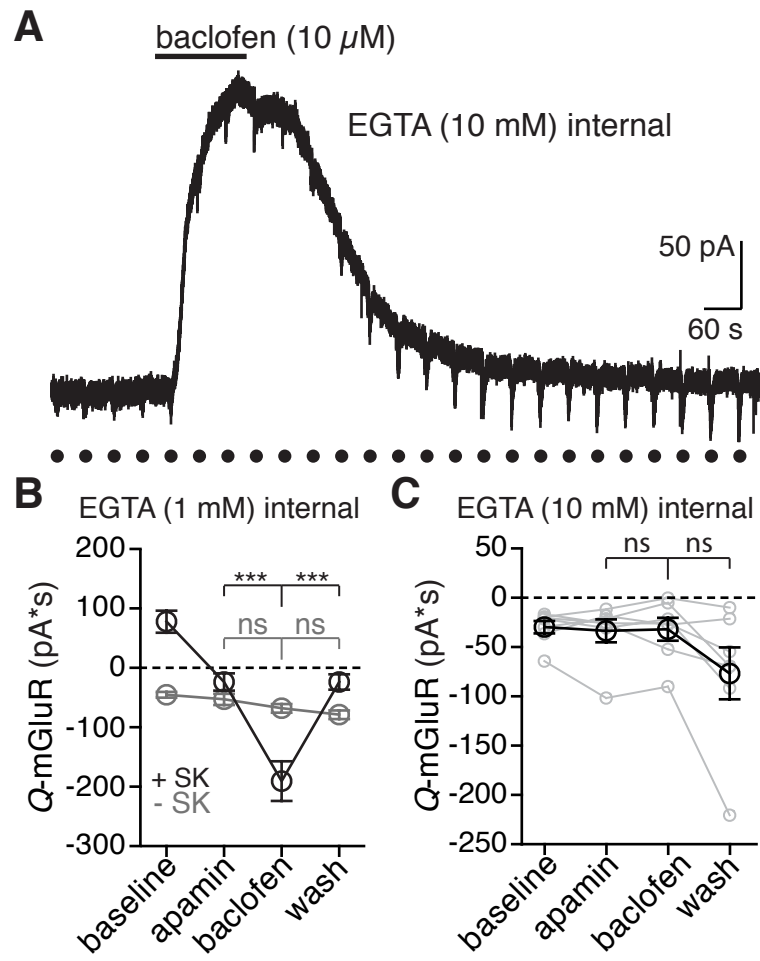


Figure S6 – Related to Figure 4. EGTA titration effects on calcium release from stores mediated currents.

A. Representative trace from a voltage clamp recording where the internal solution contained EGTA (10 mM). There was no baseline SK current, and application of baclofen (10 μ M) did not change the current produced by aspartate iontophoresis. Black circles represent the activation of mGluRs with aspartate iontophoresis.

B. Data collected with EGTA (1 mM) in the internal solution, grouped according to the presence of absence of an SK current before apamin application. If there was an SK current (black) then mGluR activation inhibited the GIRK. If there was no SK current (grey) then mGluR activation did not inhibit the GIRK. Statistics were performed with a two-way repeated measures ANOVA followed by Bonferroni, *** $p < 0.001$, ns = not significant $p > 0.99$, $n = 4$ cells for + SK, 6 cells for - SK.

C. Grouped data ($n = 8$ cells) for experiments with EGTA (10 mM) in the internal solution. Statistics were performed with a repeated measures one-way ANOVA, ns = not significant $p = 0.98$ (apamin – baclofen), $p = 0.13$ (baclofen – wash).

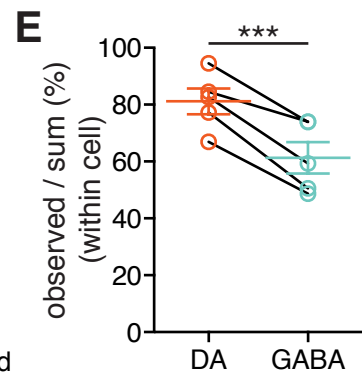
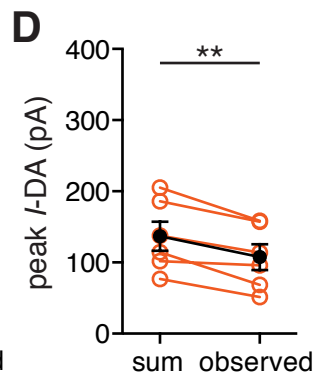
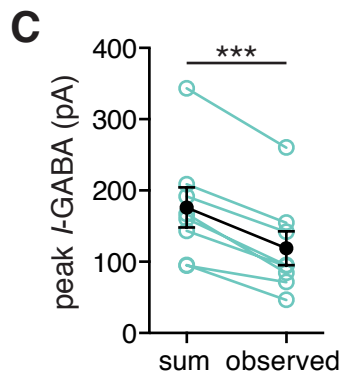
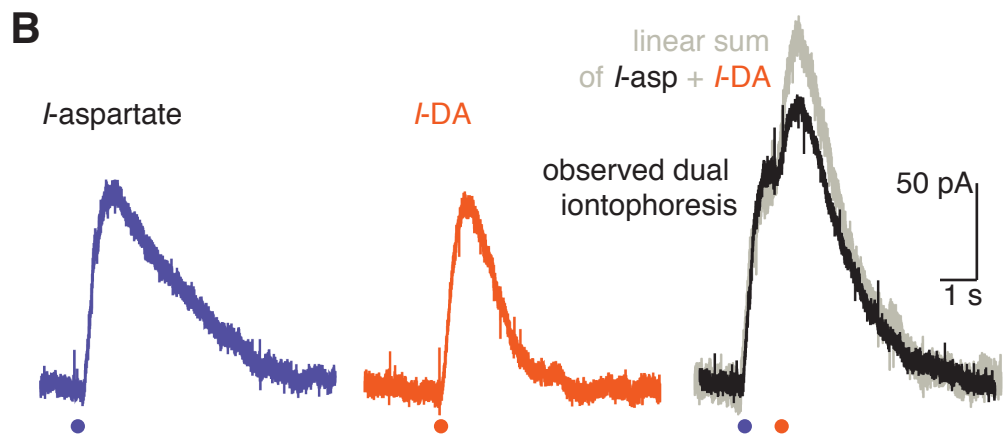
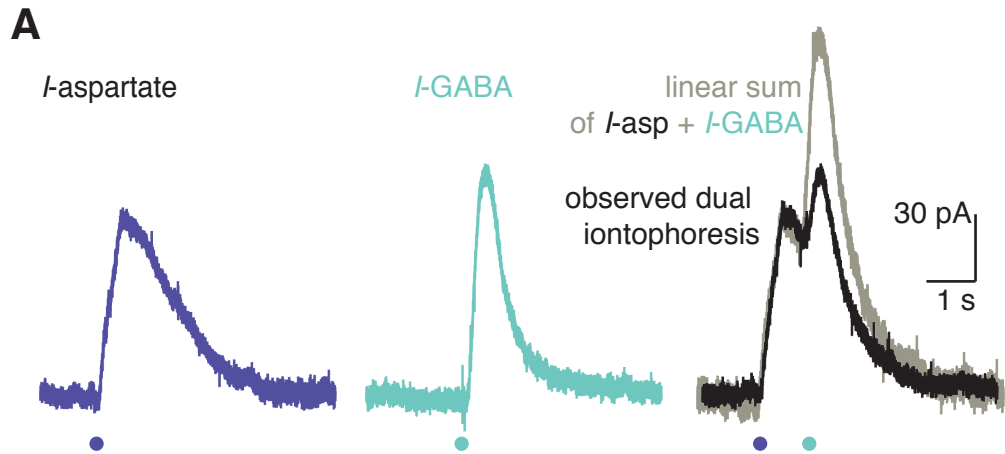


Figure S7 – Related to Figure 7. Blocking SK was not required for inhibition of GIRK by mGluR activation.

A. Representative traces from a whole-cell recording showing an example of the current mediated by iontophoresis of aspartate (purple, *left*), GABA (green, *middle*), and aspartate one second before GABA (black, *right*). Also shown is the summation of the currents produced by iontophoresis of aspartate and of GABA (grey, *right*).

B. Representative traces from a whole cell recording showing an example of the current mediated by iontophoresis of aspartate (purple, *left*), dopamine (orange, *middle*), and aspartate one second before dopamine (black, *right*). Also shown is the linear summation of the currents produced by iontophoresis of aspartate and of dopamine (grey, *right*).

C. Grouped data showing the difference between the peak of the GABA_BR GIRK current expected by linear summation versus observed experimentally. Green circles and connecting lines represent data from single cells, black points represent means and error bars s.e.m. Statistics were performed with a paired two-tailed t-test, *** $p < 0.001$, $n = 8$ cells.

D. Grouped data showing the difference between the peak of the dopamine GIRK current expected by linear summation versus observed experimentally. Orange circles and connecting lines represent data from single cells, black points represent means and error bars s.e.m. Statistics were performed with a paired two-tailed t-test, ** $p < 0.01$, $n = 6$ cells.

E. Grouped data showing the experimentally observed peak GIRK current / summated peak based on linear summation for dopamine (DA, orange) and GABA (green). Orange and green circles and connecting lines represent data from single cells, black points represent means and error bars s.e.m. Data for each pair were collected from single cells. Statistics were performed with a paired two-tailed t-test, ** $p < 0.01$, $n = 5$ cells.